

TABLE II. Observed and predicted numbers of ν_μ , ν_e and Low-energy ν_e events in the neutrino beam, and $\bar{\nu}_\mu$ and $\bar{\nu}_e$ events in the antineutrino beam. The low PID, high PID, and Peripheral samples have been combined in the ν_e ($\bar{\nu}_e$) columns, while the Low-energy sample is shown separately due to its novel status in this analysis. WS is considered signal in the ν_μ and $\bar{\nu}_\mu$ samples, but background in the ν_e and $\bar{\nu}_e$ samples. We denote as Others the sum of all remaining oscillation channels that are not listed explicitly in this table. The best-fit point for the prediction is extracted from a frequentist fit to the data with the Daya Bay 1D constraint on $\sin^2 2\theta_{13}$ [61].

Sample	Neutrino Beam			Antineutrino Beam	
	ν_μ	ν_e	Low-energy	$\bar{\nu}_\mu$	$\bar{\nu}_e$
$\nu_\mu \rightarrow \nu_\mu$	372.3	4.3	0.3	24.4	0.2
$\bar{\nu}_\mu \rightarrow \bar{\nu}_\mu$	24.5	0.1	0.0	71.5	0.2
$\nu_\mu \rightarrow \nu_e$	0.4	125.3	3.4	0.0	2.1
$\bar{\nu}_\mu \rightarrow \bar{\nu}_e$	0.0	1.8	0.1	0.0	18.9
Beam $\nu_e + \bar{\nu}_e$	0.1	26.1	0.8	0.0	6.5
NC	5.5	16.8	5.3	0.8	2.0
Cosmic	4.4	5.5	0.5	0.7	1.1
Others	1.5	0.8	0.1	0.2	0.1
Signal	$397.6^{+26.8}_{-24.2}$	$125.3^{+5.1}_{-5.1}$	$3.4^{+1.0}_{-0.8}$	$96.0^{+5.2}_{-5.7}$	$18.9^{+0.8}_{-0.9}$
Background	$11.0^{+1.3}_{-1.2}$	$55.4^{+2.6}_{-1.7}$	$7.1^{+1.3}_{-1.1}$	$1.7^{+0.2}_{-0.2}$	$12.2^{+0.5}_{-0.7}$
Predicted	408.6	180.7	10.5	97.7	31.1
Observed	384	169	12	106	32

10^{-3} eV^2 in the normal MO or $\Delta m_{32}^2 = -2.479^{+0.036}_{-0.036} \times 10^{-3} \text{ eV}^2$ in the inverted MO. The highest posterior density (HPD) for δ_{CP} is $0.93^{+0.21}_{-0.89} \pi$ in the normal MO. For both MO, the data favor regions of parameter space close to maximal mixing, $\sin^2 \theta_{23} = 0.55^{+0.02}_{-0.06}$, with no significant preference for octant. Fig. 2 shows the 90% credible regions in the $\Delta m_{32}^2 - \sin^2 \theta_{23}$ plane extracted from this analysis compared to other experiments and previous NOvA results. Good agreement is seen across all the experiments. The goodness of fit is estimated using posterior-predictive p -values (PPP), where sampled models that describe the data well will have values close to 0.5 [75]. With the 1D Daya Bay constraint on θ_{13} , the total PPP across all data samples is 0.48, indicating good agreement between the data and the model. The PPP does not change significantly when either no external constraint on θ_{13} is used or when the two-dimensional $\sin^2 2\theta_{13} - \Delta m_{32}^2$ constraint from Daya Bay is applied [62].

The NOvA data lie in the degenerate region of parameter space which can be accommodated within either MO by varying the preferred values of δ_{CP} , as shown in Fig. 3. Bayes factors, representing ratios of posterior probabilities between two hypotheses, are used here to represent how much more probable one model is over another. Bayes factors for the preference of CP violation

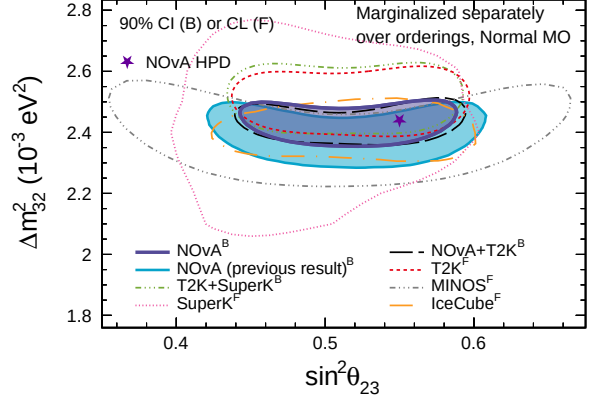


FIG. 2. Comparisons of the 90% intervals for $\Delta m_{32}^2 - \sin^2 \theta_{23}$ in the normal MO with NOvA 2020 results and superimposed contours from other experiments [76–80], including the 2024 joint NOvA-T2K analysis [81]. The NOvA results are with the 1D Daya Bay constraint on θ_{13} applied [61]. Contours labeled B are from Bayesian analyses, while those labeled F are from frequentist analyses, used when Bayesian results were not available.

over CP conservation were extracted using the Savage-Dickey method [82, 83] based on the Jarlskog invariant [84], with Jarlskog invariant of 0 representing CP conservation. The Bayes factors for CP violation are 1.1 (containing 51% of the posterior probability) in the normal MO and 4.3 (81%) in the inverted MO, and remain compatible with both CP conservation and CP violation.

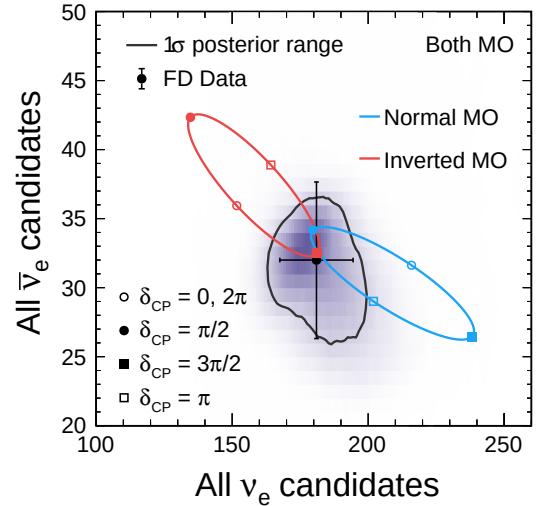


FIG. 3. Bi-event plot showing the posterior probability for the predicted number of ν_e and $\bar{\nu}_e$ events in purple, with the 1σ credible interval and the FD data point marked. The two ovals, red for the inverted MO and blue for the normal MO, show the prediction with all the parameters fixed at the NOvA best-fit, varying only δ_{CP} , with four δ_{CP} points marked.